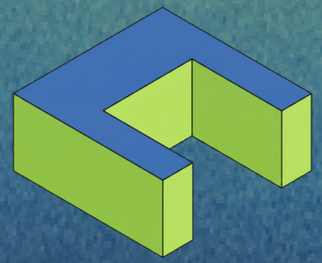


CONTEXTA

Contextual Adaptive English Platform

Project ID : COE2026-10

Presented by : Mr. Thanasud Ratanathaworn (ID: 663040650-9)
Mr. Punnawit Seangsuriyakat(ID: 663040121-6)

Supervisory Team : Asst. Prof. Dr. Manasawee Kaenampornpan
Asst. Prof. Dr. Kornchawal Chaipah
Asst. Prof. Dr. Sarun Paisarnsrisomsuk

TODAY AGENDA

01.	Background & Problem Statement	06.	Challenges & Solutions
02.	Objectives	07.	Design
03.	What we have done?	08.	Tech Stack & Tools
04.	Comparison	09.	Progress
05.	Key Features	10.	Next Steps
11.	Q&A		

BACKGROUND

English is the #1 Most Tutored Subject in Thailand.

Ref: เมื่อคุณภาพการศึกษาใช้ “คะแนนชี้วัด” ผู้ปกครองทุ่มปีละกว่าหมื่นล้านให้เด็กกวดวิชาเพื่อ “ติ๊กถูก”, THAIPUBLICA

Ref: 50 อันดับวิชาที่นักเรียนต้องการเรียนมากที่สุด, Tutor Ferry

Thai students spend more on English tutoring than any other subject (26.96% of total demand).

“The goal is communication. But the learning journey is optimized for exams”

115	Saudi Arabia	404
116	Thailand	402
116	Yemen	402

Positioned among the **lowest-ranked in Asia** for the second consecutive year

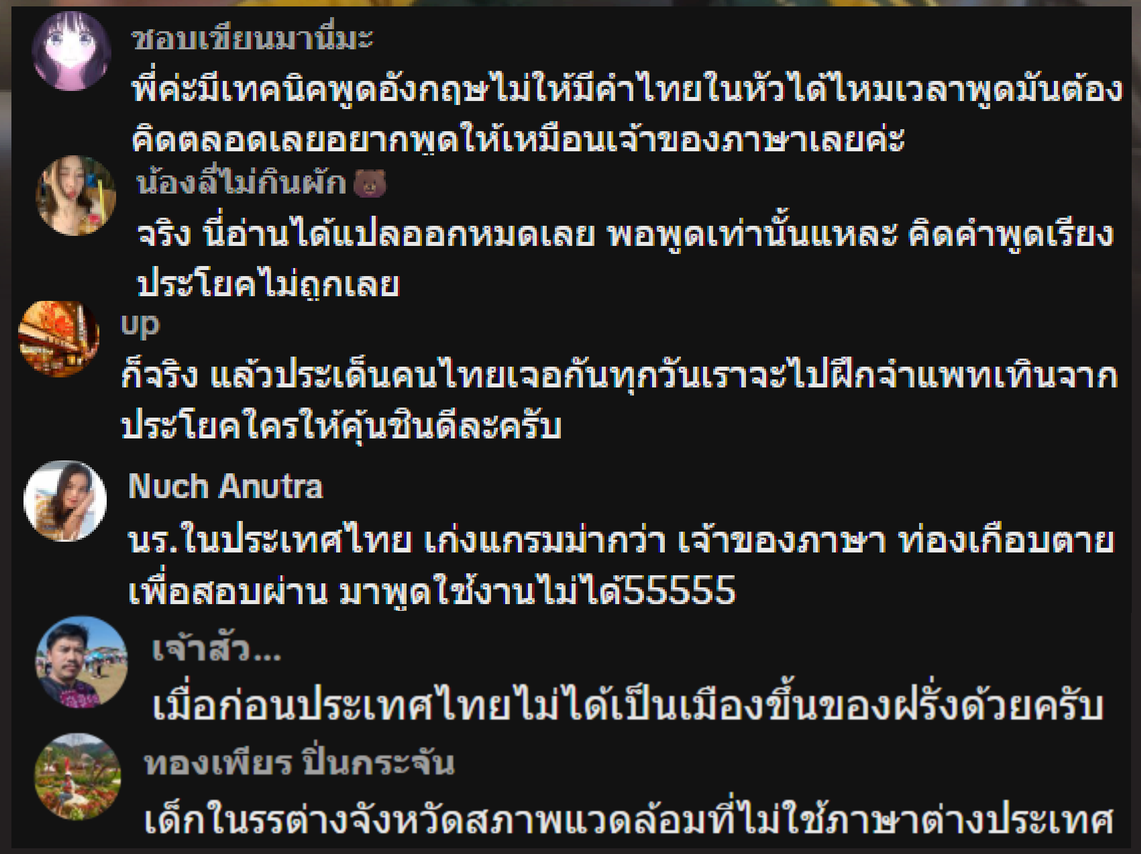
English proficiency is ranked 116th

Part speaking ranks **3rd** of 4 skills

PROBLEM STATEMENT

Fear of grammar mistakes and "Accent Shaming" from society.

Ref: kruramil, KOLs สายสอนภาษาอังกฤษ



The Technical Gaps

- **Irrelevant Context: Existing platforms use static, textbook scenarios**
  - Impact: Causes a severe disconnect, forcing rote memorization over actual understanding, leading to low engagement and high drop-out rates.
- **Native Bias: Standard ASR models strictly demand native-like accents, unfairly penalizing "Tinglish" speakers and causing performance anxiety.**
  - Impact: Unfairly penalizes "Tinglish" speakers, destroying confidence and causing performance anxiety.

*ASR = Automatic Speech Recognition

OBJECTIVES

To develop an English learning mobile application equipped with

- **Adaptive Context Lessons:** Dynamically generating personalized practice scenarios based on user data.
- **Tinglish-Tolerant ASR:** Optimized to accurately distinguish between Thai-accented speech and actual mispronunciation.
- **Hybrid Scoring System:** Prioritizing semantic meaning and intonation over native-like perfection.

*ASR = Automatic Speech Recognition

What we have done?

- **Curriculum Research & Data Preparation:** Analyzed the Thai Basic Education Core Curriculum (OBEC) and built a categorized vocabulary dataset enriched with Thai translations and CEFR levels.
- **AI PoC Training:** Curated a specialized audio dataset and successfully trained the initial Proof of Concept (PoC) for our Tinglish-Tolerant ASR model.
- **Pain Point Validation:** Validated user learning barriers (e.g., anxiety, lack of confidence) through academic literature reviews.
- **UI/UX & Core Flow:** Designed 6 core application screens (Onboarding, Dashboard, Roleplay) with adaptive learning mechanics (Guided to Free Speaking).

Comparison

Features	CONTEXTA	ELSA Speak	Duolingo Max	Yoodli
Tinglish-Tolerant ASR	✓	⚠	✗	✗
Accent vs. Mispronunciation Detection	✓	✓	⚠	✗
Hyper-Personalized Context	✓	⚠	✗	✗
Upload-to-Practice (Slides/Docs)	✓	✗	✗	⚠
Thai UI & Feedback	✓	✓	✓	✗

Key Features

1

AI Personalized Context

Creating context that truly resonates with users.

2

Tinglish Friendly AI

An AI system that understands Thai accents and stress patterns.

3

AI Upload-to-Practice

Real-World Scenario Practice Mode.

Key Features

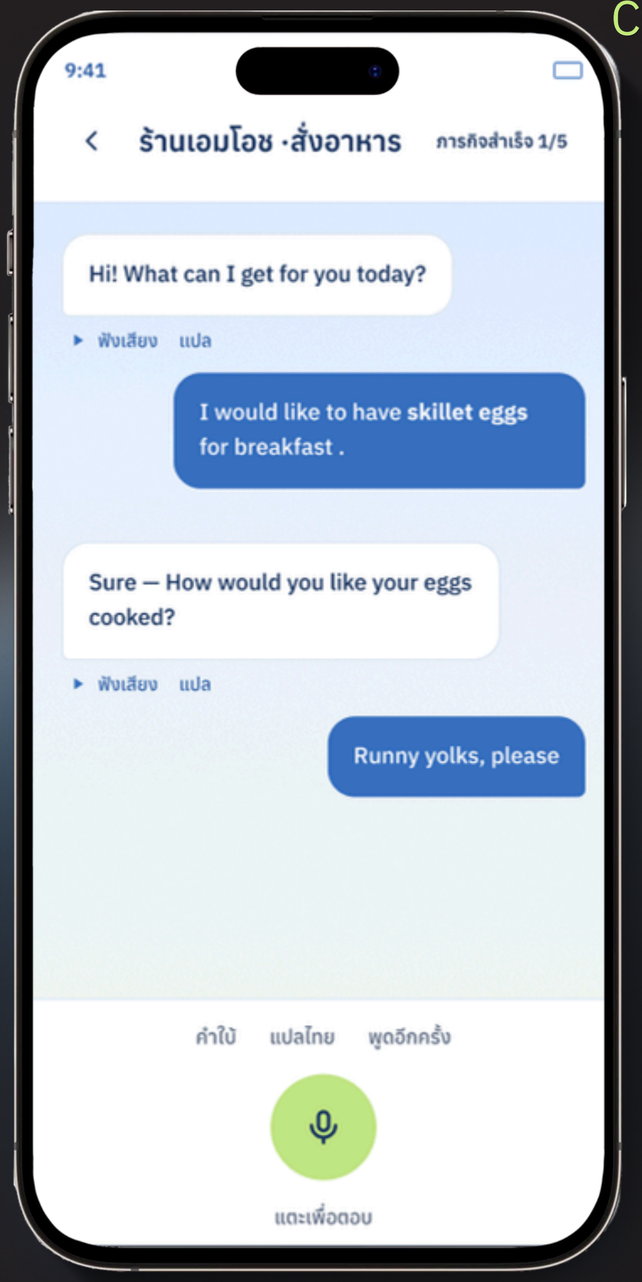
1 AI Personalized Context

Creating context that truly relate with users.

- Extracts user data (e.g., career, interests, goals)
- Generate highly relavant scenarios

Key Techniques

- Build a vocabulary database using LLMs to enrich each word across 3 dimensions: level-appropriate Thai translations, concrete/abstract categorization, and Oxford 5000 topic tags (including part of speech and CEFR level).
- Leverage AI to select topics and generate personalized exercises tailored to each learner, moving away from generic, one-size-fits-all lessons.



Key Features

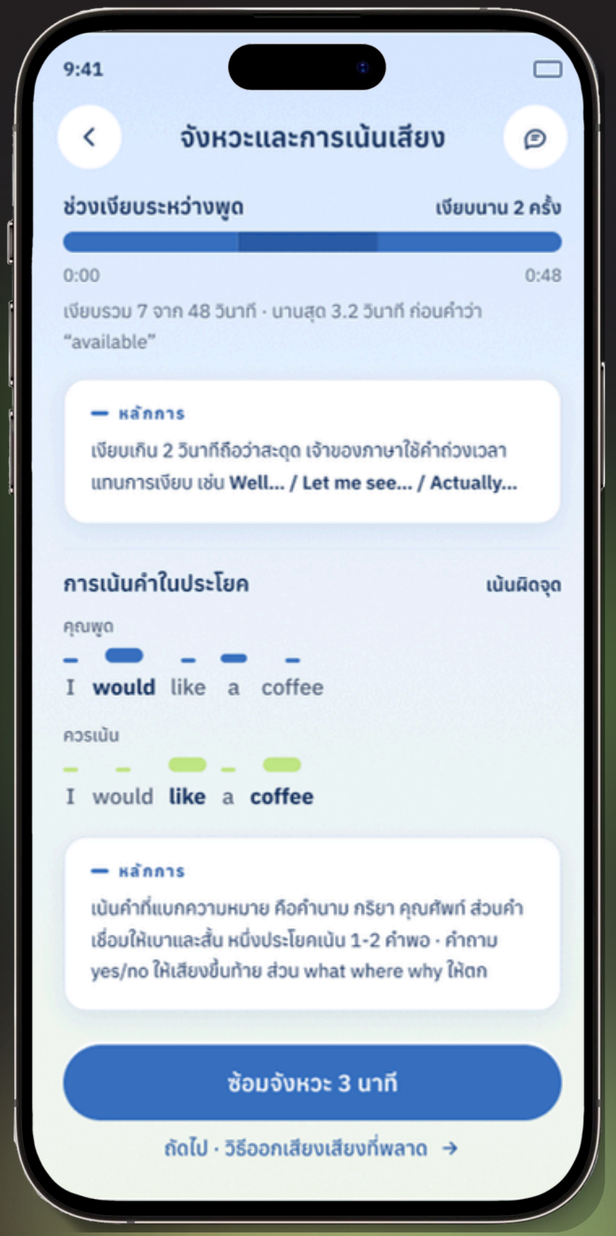
AI that understands Thai accents.

2 Tinglish Friendly AI

- The AI model is specifically trained on Thai pronunciation to best understand Thai speakers.
- No need to sound like a native speaker—focus on correct intonation to communicate effectively and get your point across.

Key Techniques

- Fine-tuning the OpenAI Whisper model using LoRA techniques to enhance Thai accent recognition.
- Curating the audio dataset to establish benchmarks for distinguishing between a 'Thai accent' and actual 'mispronunciation.'
- Evaluating model performance using unseen sentences and unseen speakers.



Key Features

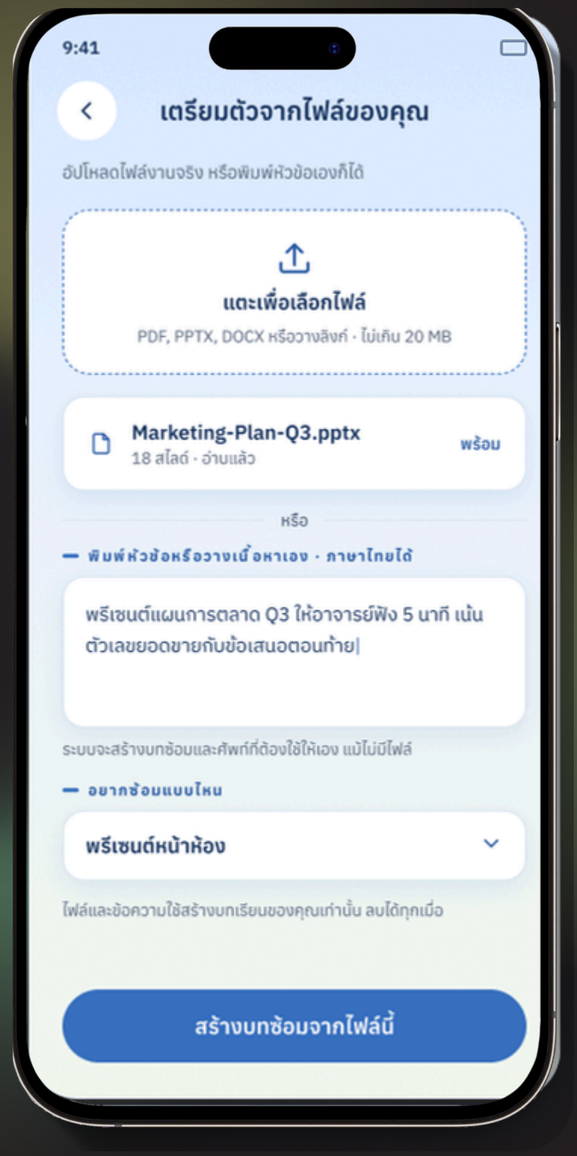
Real-World Scenario Practice Mode

3 AI Upload-to-Practice

- Working professionals and students often already have specific content they need to present.
- Transforming a standard language learning app into a personal tutor—a feature no competitor in the market currently offers.

Key Techniques

- Document Parsing: Supports PDF, PPTX, and DOCX uploads.
- LLM Script Generation: Converts raw bullet points into natural, conversational speaking scripts.
- Voice Processing Pipeline: Integrates Whisper LoRA (ASR) + VAD (Voice Activity Detection).
- Hybrid Evaluation: Scores based on Semantic Similarity and meaning, rather than exact-word matching.



CHALLENGES & SOLUTIONS

CHALLENGES & SOLUTIONS

Pain Point Validation

- Challenge: Initial pain points lacked academic evidence.
- Solution: Conducted literature reviews to prove that personalized context significantly boosts learning motivation.

The Curriculum Gap (OBEC vs. CEFR)

- Challenge: Thai education (OBEC) lacks distinct CEFR levels for in-between grades.
- Solution: Designed a custom Sub-leveling System, mapping OBEC vocabulary quotas to precise CEFR standards (Pre-A1 to C2).

Vocabulary Data Limitations

- Challenge: Open-source CEFR datasets lack Thai translations and proper categorizations.
- Solution: Built an automated Data Enrichment Pipeline with a 3-layer tagging system to categorize them (CEFR, Concept, Topic).

AI Model Overfitting (PoC)

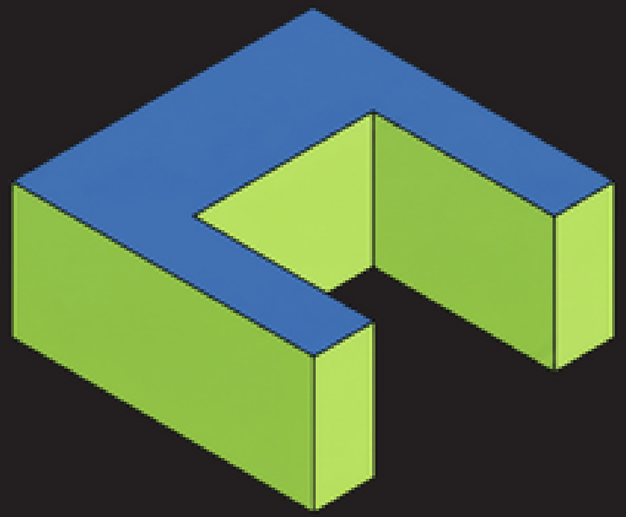
- Challenge: Initial Whisper fine-tuning hit 0% WER, indicating severe overfitting.
- Solution: Restructured the dataset to include Unseen Speakers and Unseen Sentences for robust evaluation.

*PoC = Proof of Concept
*OBEC = Office of the Basic Education Commission

CONTEXTA

DESIGN

DESIGN LOGO



THE NAME

CONTEXTA

CONTEXT —REAL SITUATIONS,

TA — A TEACHING ASSISTANT

Primary	second
Blue #3770c1	Green (#c3e97c)

CONTEXTA

TECH STACK AND TOOLS

TECH STACK

FRONTEND

BACKEND

SPEECH AI

OpenAI Whisper

DATABASE

PostgreSQL

CLOUD

LLM

TOOL

DESIGN

API

AI

CONTEXTA

PROGRESS

PROGRESS

VOCABULARY & CONTENT DATA

50%

- Vocabulary database built and enriched by LLM
- Level-appropriate Thai translations + concrete / abstract tags
- Oxford 5000 topics mapped with part of speech and CEFR level

THAI-ACCENT ASR

0%

- Audio dataset curated for accent vs. mispronunciation
- Whisper fine-tuned with LoRA on Thai-accented speech

PRODUCT & UI/UX MOBILE

50%

- Core practice loop designed — record, score, drill
- Hi-fi screens done for pacing and stress feedback

UPLOAD-TO-PRACTICE

0%

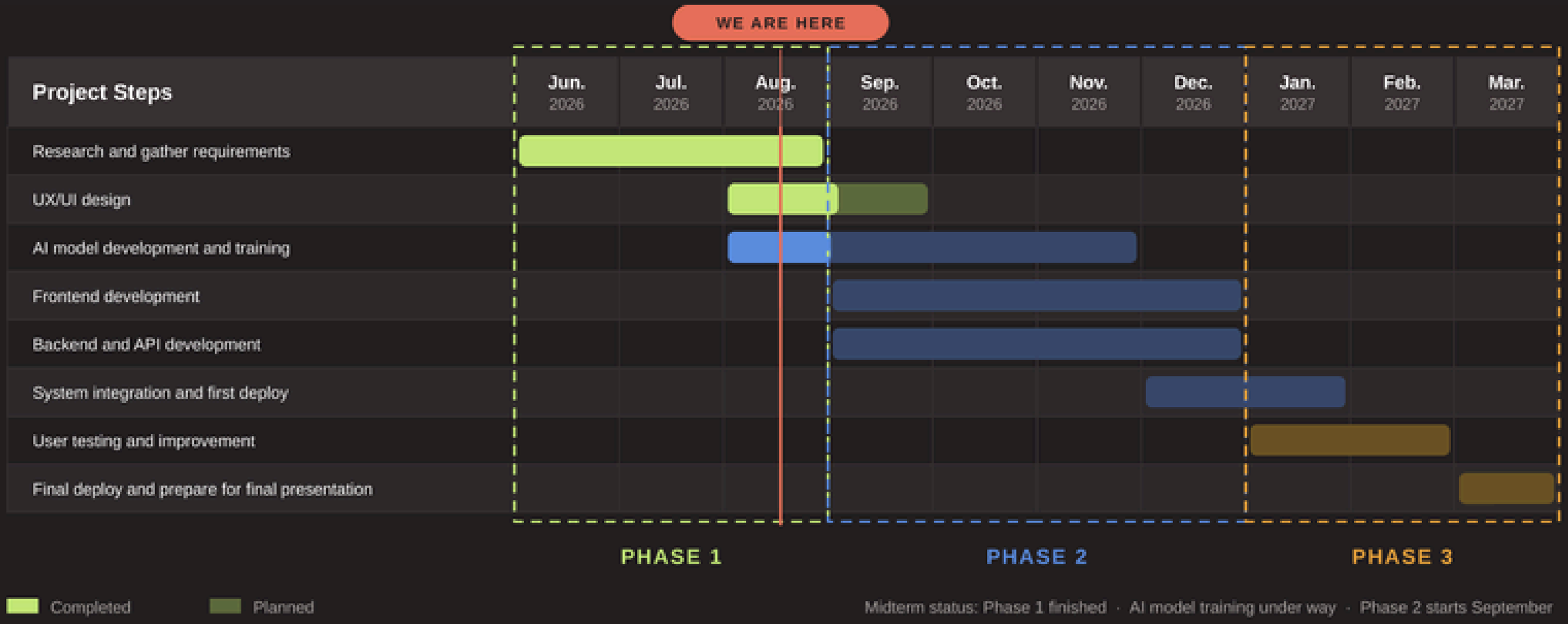
- Deck parsing for PDF / PPTX / DOCX
- LLM turns slide bullets into a natural speaking script

CONTEXTA

NEXT STEPS

NEXT STEPS

WORK EVERY THURSDAY 1:00 P.M - 4: 00 P.M



Thai people aren't bad at English.
We were just forced to learn it in a
"context" we never actually use.

Q&A

CONTEXTA

BACK UP SLIDES

```
!python -m src.eval_wer
```

```
[transformers] Ignoring clean_up_tokenization_spaces=True for BPE tokenizer WhisperTokenizer. The clean_up_tokenization post-processing step is designed for WordPiece tokenizers and is destruc
True text : Please call Stella. Ask her to bring these things with her from the store: Six spoons of fresh snow peas, five thick slabs of blue cheese, and maybe a snack for her brother Bob. We
Base Pred : Please call Stella, ask her to bring these things with her from the store. Six spoons of fresh snow peas, five thick slabs of blue cheese and maybe a snack for her brother Bob. We
LoRA Pred : Please call Stella. Ask her to bring these things with her from the store, Six spoons of fresh snow peas, five thick slabs of blue cheese, and maybe a snack for her brother Bob. We
----------------------------------------
True text : Please call Stella. Ask her to bring these things with her from the store: Six spoons of fresh snow peas, five thick slabs of blue cheese, and maybe a snack for her brother Bob.
Base Pred : Peace call Stella asked her to bring these things with her from the store. 6 spoons of fresh snow peas, 5 thick slabs of blue cheese and maybe a snack for her brother Bob.
LoRA Pred : Please call Stella. Ask her to bring these things with her from the store, Six spoons of fresh snow peas, five thick slabs of blue cheese, and maybe a snack for her brother Bob.
----------------------------------------
True text : We also need a small plastic snake and a big toy frog for the kids. She can scoop these things into three red bags, and we will go meet her Wednesday at the train station.
Base Pred : We also need a small plastic snack and a big toy fork for the kids. She can scoop this thing into three legs bags and we will go meet her Wednesday at the train station.
LoRA Pred : We also need a small plastic snake and a big toy frog for the kids. She can scoop these things into three red bags, and we will go meet her Wednesday at the train station.
----------------------------------------
True text : Please call Stella. Ask her to bring these things with her from the store: Six spoons of fresh snow peas, five thick slabs of blue cheese, and maybe a snack for her brother Bob.
Base Pred : Please call Stella. Ask her to bring this thing with her from the store. 6 spoons of flasno peas, 5 thick slabs of blue cheese and maybe a snack
LoRA Pred : Please call Stella. Ask her to bring these things with her from the store, Six spoons of fresh snow peas, five thick slabs of blue cheese, and maybe a snack for her brother Bob.
----------------------------------------
True text : We also need a small plastic snake and a big toy frog for the kids. She can scoop these things into three red bags, and we will go meet her Wednesday at the train station.
Base Pred : for her brother Bob. We also need a small plastic snake and a big toy flock for the kids. She can scoop this thing into three-let bag and we will go meet her Wednesday at the trair
LoRA Pred : We also need a small plastic snake and a big toy frog for the kids. She can scoop these things into three red bags, and we will go meet her Wednesday at the train station.
----------------------------------------
True text : That restaurant serves the best noodles in town.
Base Pred : That restaurant served the best noodles in town.
LoRA Pred : That restaurant served the best noodles in town.
----------------------------------------
True text : We should practice speaking more often.
Base Pred : We should practice speaking more often.
LoRA Pred : We chose practice speaking more often.
----------------------------------------
True text : The train was late because of the rain.
Base Pred : The train was late because of the rain.
LoRA Pred : The train was late because of the rain.
----------------------------------------
True text : I'm trying to improve my pronunciation this year.
Base Pred : I'm trying to improve my pronounciation this year.
LoRA Pred : I'm trying to improve my pronounation this year.
----------------------------------------
True text : Thank you very much for your help today.
Base Pred : Thank you very much for your help today.
LoRA Pred : Thank you very much for your help today.
----------------------------------------
```

```
----------------------------------------
True text : We should practice speaking more often.
Base Pred : We should practice speaking more often.
LoRA Pred : We should practice speaking more often.
----------------------------------------
True text : The train was late because of the rain.
Base Pred : Detain was late because of the lane.
LoRA Pred : The train was late because of the rain.
----------------------------------------
True text : I'm trying to improve my pronunciation this year.
Base Pred : I'm trying to improve my pronunciation this year.
LoRA Pred : I'm trying to improve my pronunciation this year.
----------------------------------------
True text : Thank you very much for your help today.
Base Pred : Thank you very much for your help today.
LoRA Pred : Thank you very much for your help today.
----------------------------------------
True text : That restaurant serves the best noodles in town.
Base Pred : That's the salon served the best noodle in the town
LoRA Pred : That restaurant served the best noodle in the town.
----------------------------------------
True text : We should practice speaking more often.
Base Pred : We should practice speaking more authentically.
LoRA Pred : We should practice speaking more often.
----------------------------------------
True text : The train was late because of the rain.
Base Pred : The 10W because of the rim
LoRA Pred : The train was late because of the rain.
----------------------------------------
True text : I'm trying to improve my pronunciation this year.
Base Pred : I'm trying to improve my pronunciation this year.
LoRA Pred : I'm trying to improve my pronunciation this year.
----------------------------------------
True text : Thank you very much for your help today.
Base Pred : Thank you very much for your help today.
LoRA Pred : Thank you very much for your help today.
----------------------------------------
```

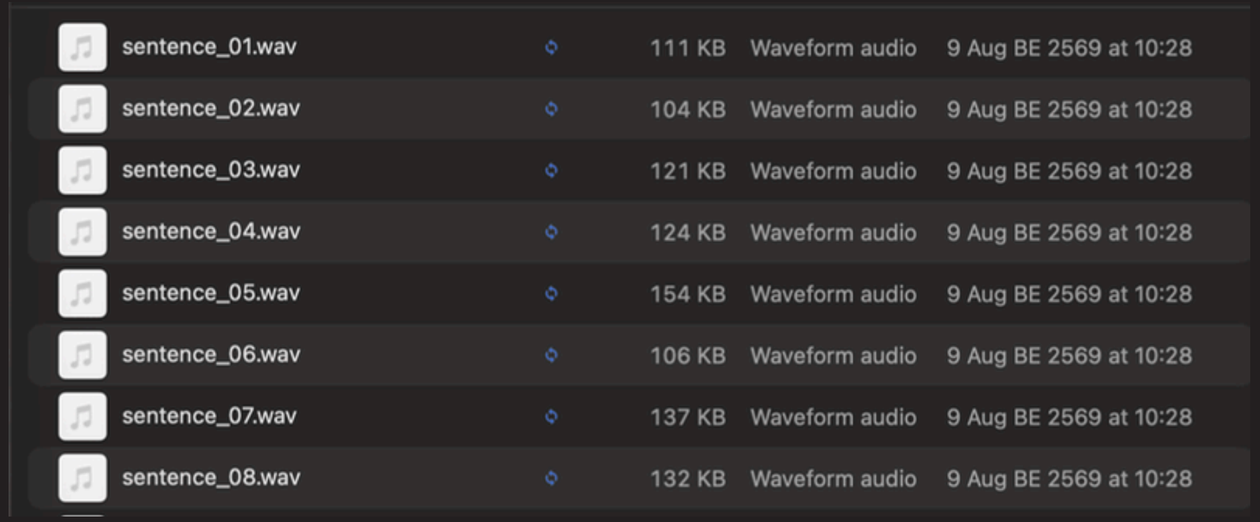
```
=== unseen_sentence (n=15) ===
Base Whisper WER: 14.91%
LoRA Whisper WER: 6.14%

=== unseen_speaker (n=5) ===
Base Whisper WER: 14.49%
LoRA Whisper WER: 0.00%

=== overall (n=20) ===
Base Whisper WER: 14.64%
LoRA Whisper WER: 2.18%
========================================
```

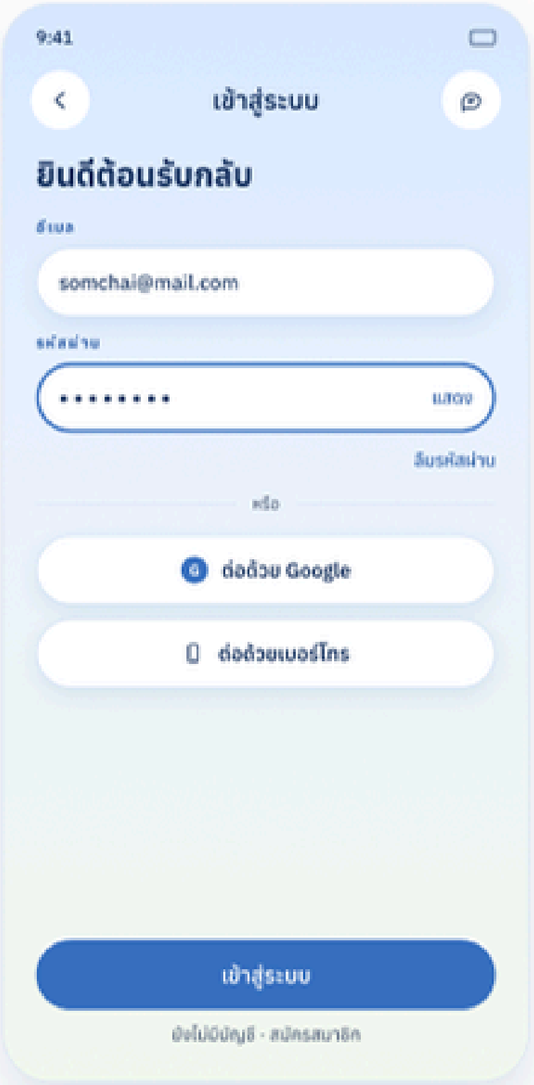
```
==================================================
SENTENCES TO READ
==================================================

01. The cat sat on the desk.
02. Please stop at the next big shop.
03. I need to read this book about the world.
04. She asked her friends to help.
05. We walked down the street to the station.
06. Can you collect the tickets for the trip?
07. I really think this is the right answer.
08. Three of these things belong to my brother.
09. I would like to think about that first.
10. This is the best day of my life.
11. He bought a light blue jacket yesterday.
12. The ship arrived before the storm started.
13. Put the sheep back in the field.
14. My schedule is full for the whole month.
15. Don't forget to lock the door before you leave.
16. The teacher asked us to finish the test.
17. I want to visit Japan next December.
18. Traffic was heavy near the airport this morning.
19. She works at a hospital in Bangkok.
20. Can we talk about the project tomorrow?
21. The coffee shop closes at eight tonight.
22. My favorite subject in school was biology.
23. He plays guitar in a small band.
24. The weather report said it will rain today.
25. I usually study English after dinner.
26. That restaurant serves the best noodles in town.
27. We should practice speaking more often.
28. The train was late because of the rain.
29. I'm trying to improve my pronunciation this year.
30. Thank you very much for your help today.
```



0 เข้าใช้งาน · สมัครและยืนยันตัวตน

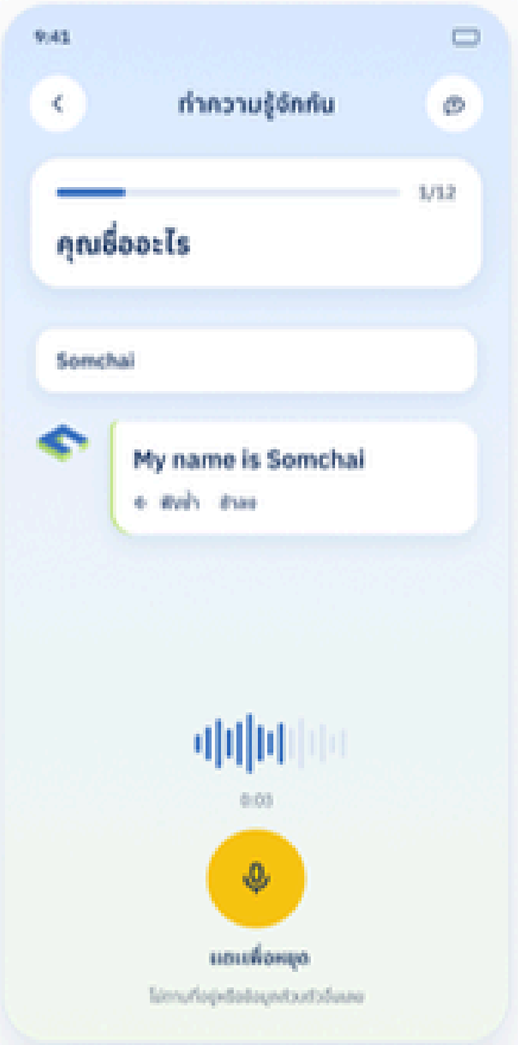
0.1 หน้าต้อนรับ
ออกคุณค่าก่อนขอให้สมัคร

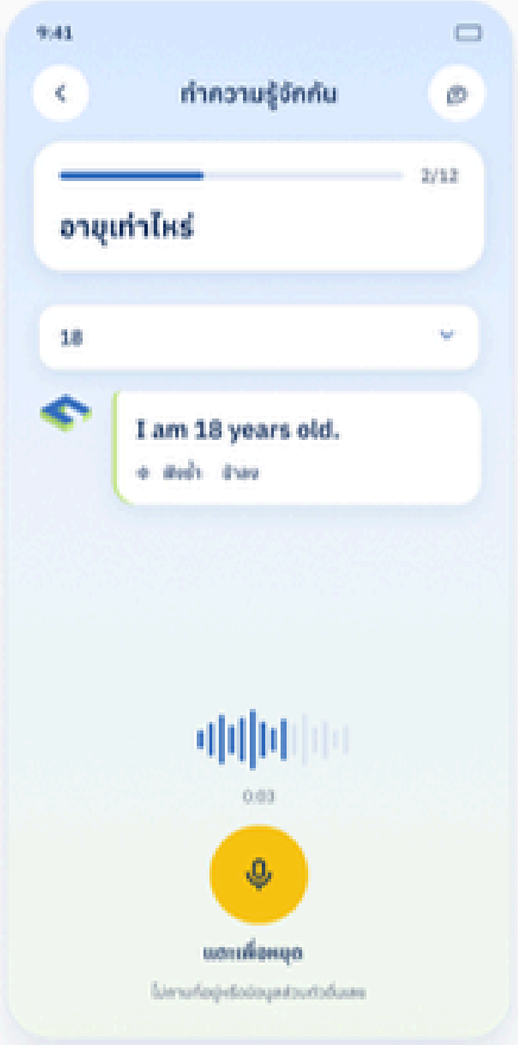


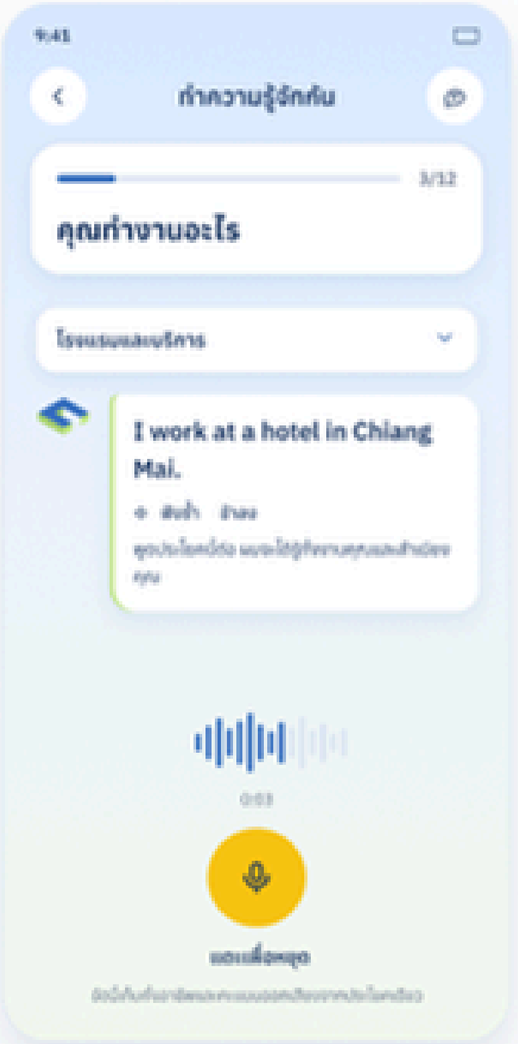
0.2 เข้าสู่ระบบ
อีเมล รหัสผ่าน หรือ Google และเบอร์โทร

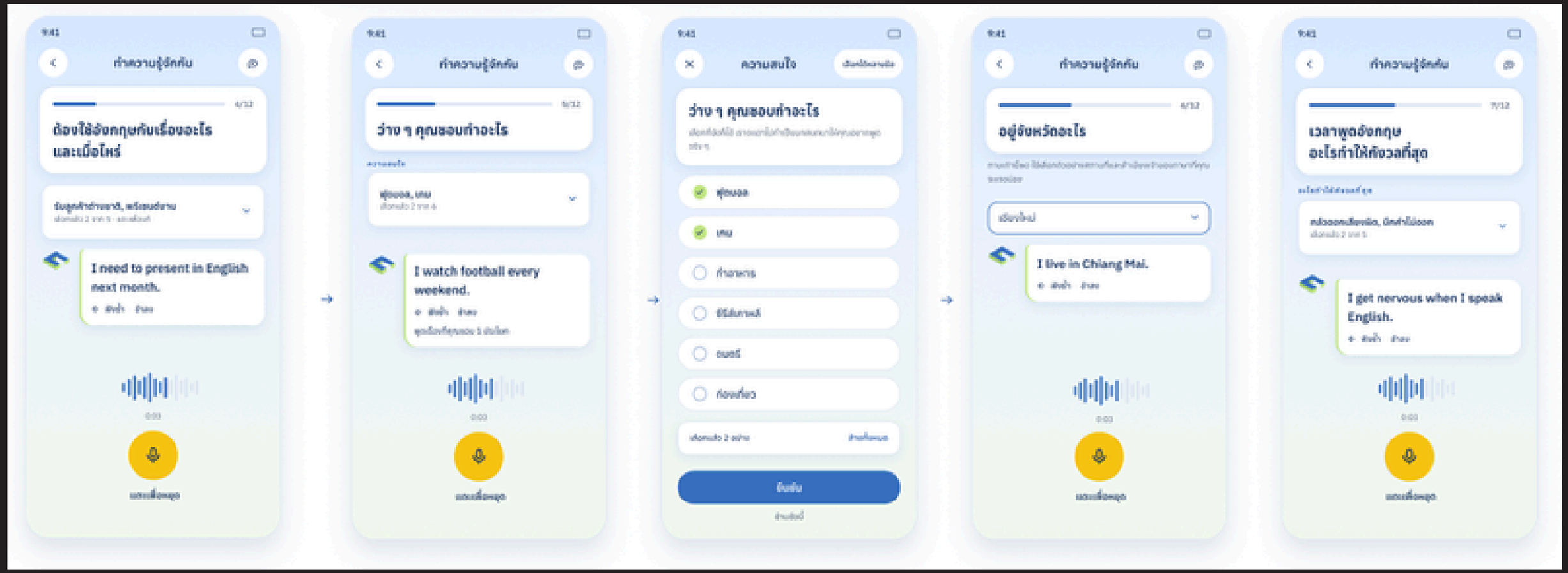
0.3 ยืนยัน OTP
ยืนยันแล้วเข้าสู่บันที่ 1

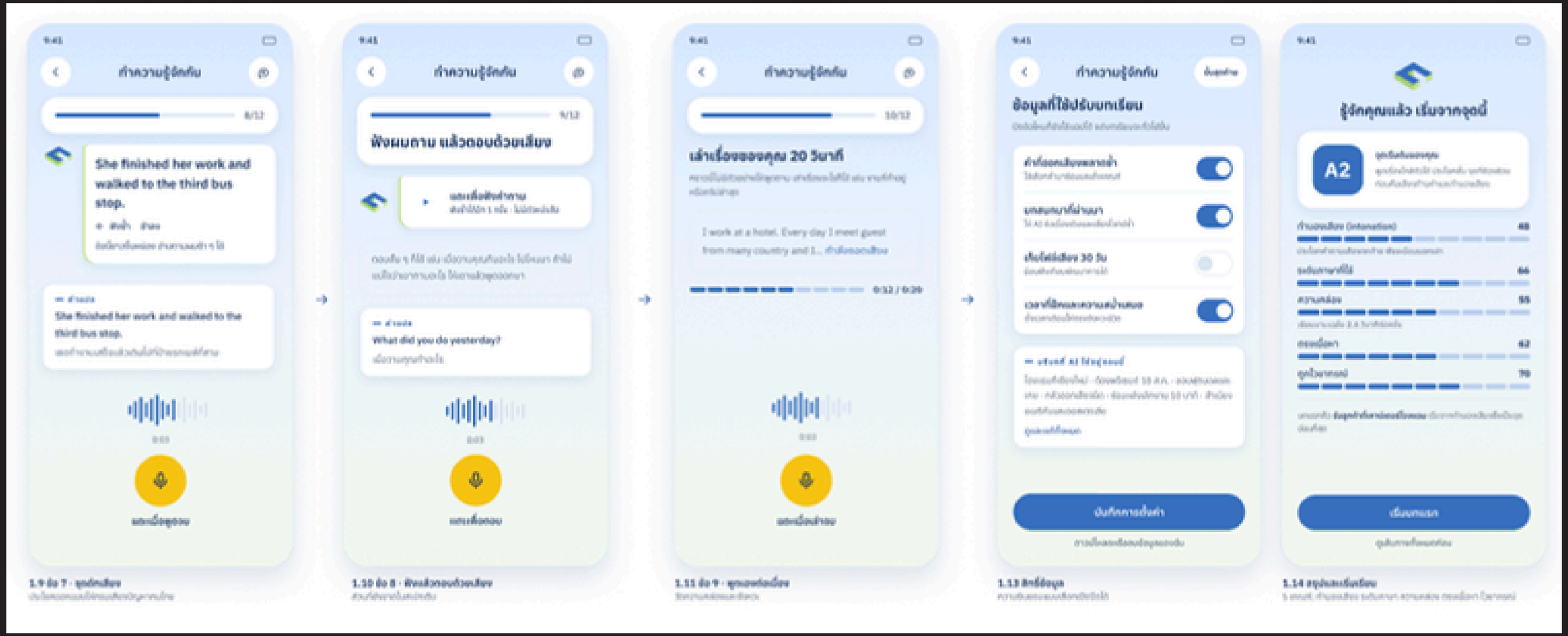
1 Onboarding A+B · พูดทุกข้อ ไม่มีคำว่าสอบ








9:41
ทำความรู้จักกัน
4/12
ต้องใช้อังกฤษกับเรื่องอะไร และเมื่อไหร่
I need to present in English next month.
9:41
ทำความรู้จักกัน
5/12
ว่าง ๆ คุณชอบทำอะไร
I watch football every weekend.
9:41
ความสนใจ
ว่าง ๆ คุณชอบทำอะไร
ฟุตบอล
เกม
ทำอาหาร
ดีไซน์ภาพ
ดนตรี
ท่องเที่ยว
9:41
ทำความรู้จักกัน
6/12
อยู่จังหวัดอะไร
I live in Chiang Mai.
9:41
ทำความรู้จักกัน
7/12
เวลาพูดอังกฤษ อะไรทำให้กังวลที่สุด
I get nervous when I speak English.



1.9 ช่อง 7 · จุดเริ่มเสียง
ประโยคตอบแรกที่ต้องเสียงดังคุยกันก่อน

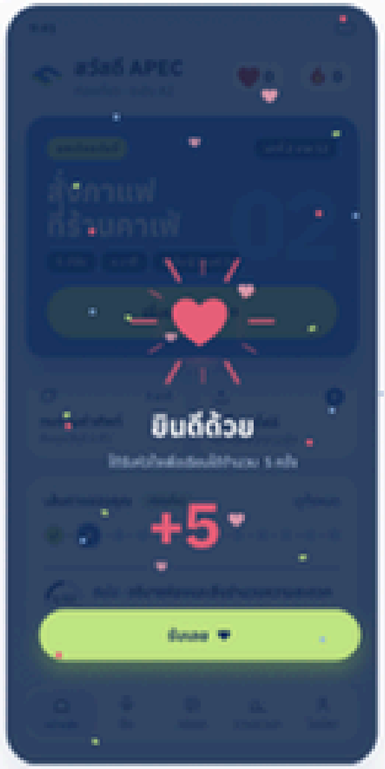
1.10 ช่อง 8 · ฟังแล้วตอบด้วยเสียง
สำหรับฝึกการโต้ตอบกลับ

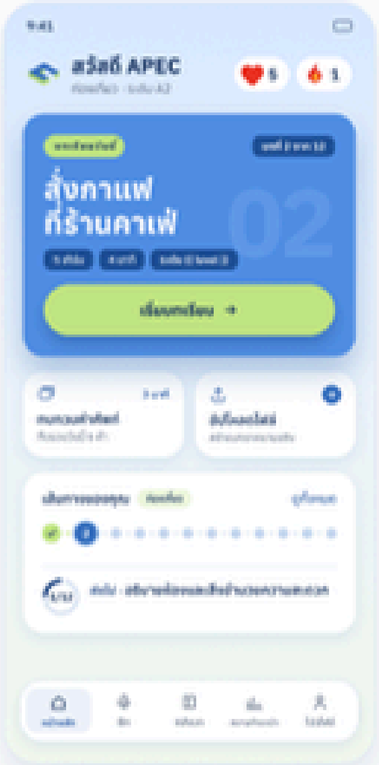
1.11 ช่อง 9 · พูดนอกห้องเรียน
ฝึกการพูดนอกห้องเรียน

1.13 ฝึกซ้ำข้อมูล
ความชัดของระบบที่ใช้ปรับให้

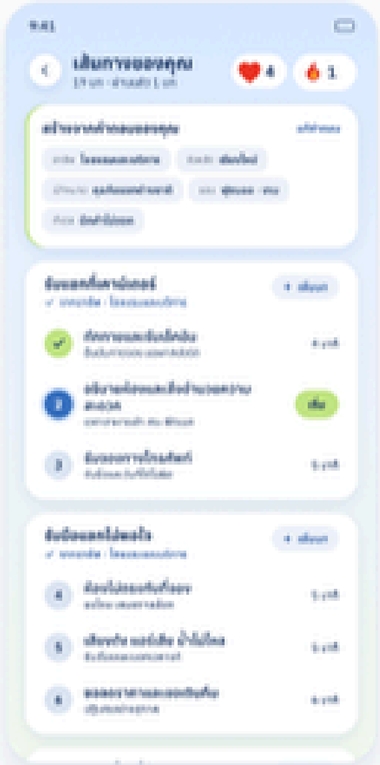
1.14 สรุปและเริ่มเรียน
ระดับ: ทำของเสีย ระดับความ คำถามสั้นๆ ประโยค

2 หน้าหลักและการเลือกสถานการณ์

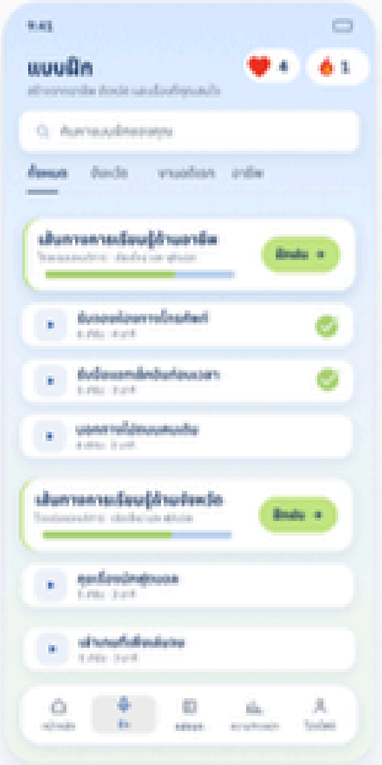




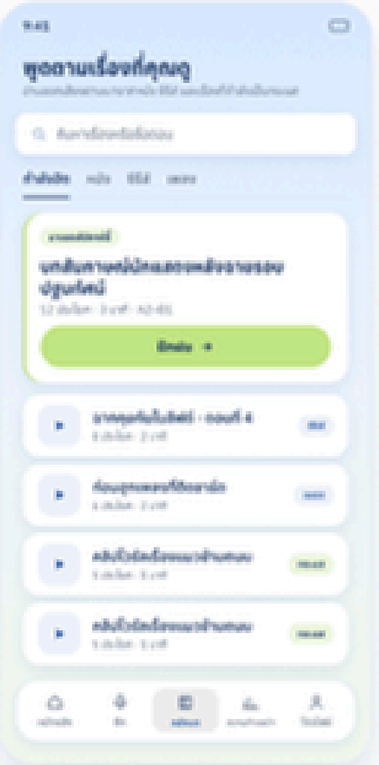
2.1 หน้าหลัก - สังคีตเวอร์ชันนี้



2.2 รายการเกมเลือกอันดับสถานการณ์



2.3 แบบฝึก

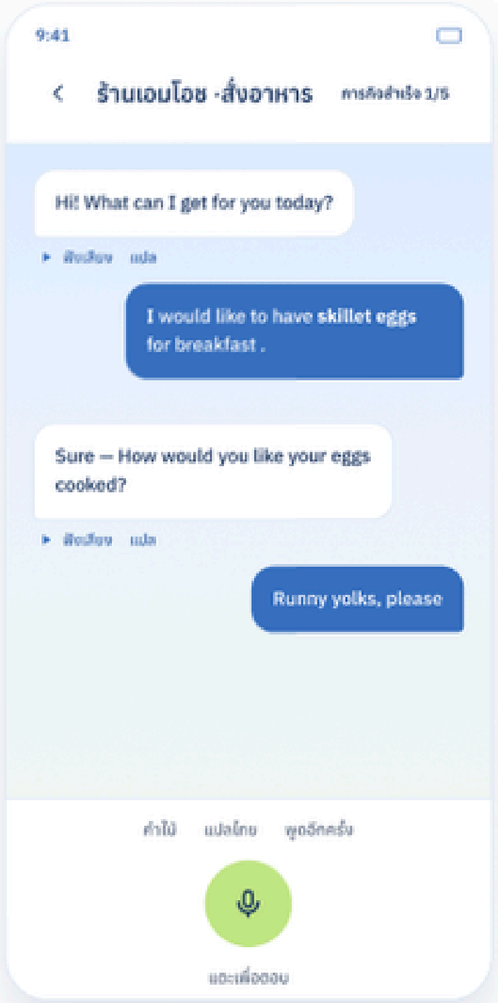


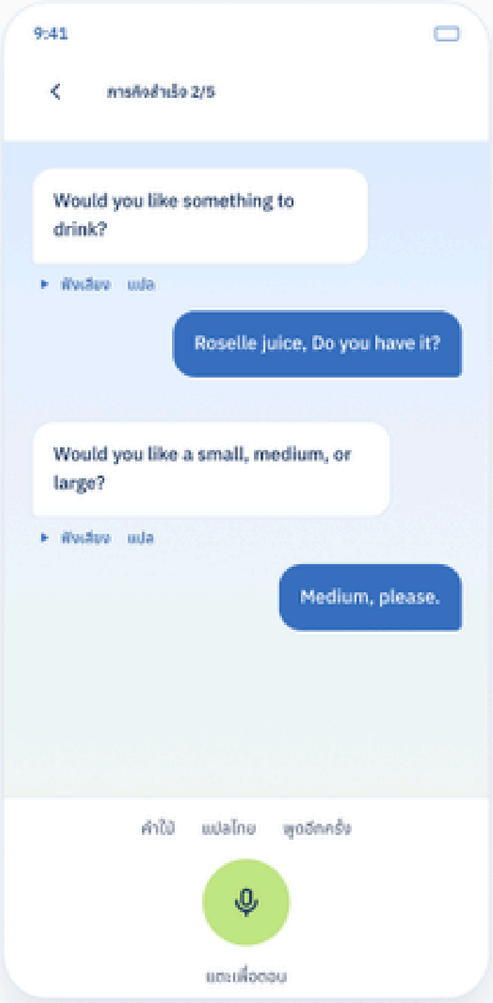
2.4 ชุดสถานเรื่องที่คุณดู

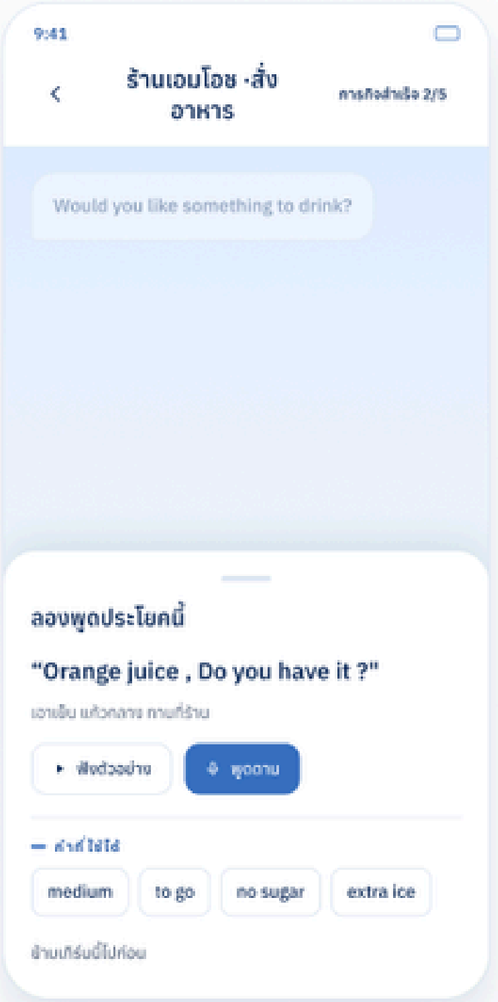
2.5 อัปโหลดไฟล์ หรือพิมพ์เอง
ไฟล์งานจริง หรือพิมพ์หัวข้อเป็นภาษาไทย → บทซ้อม

2.6 บทที่ระบบสร้างให้
แบ่งเป็นช่วง + ศัพท์เฉพาะจากไฟล์

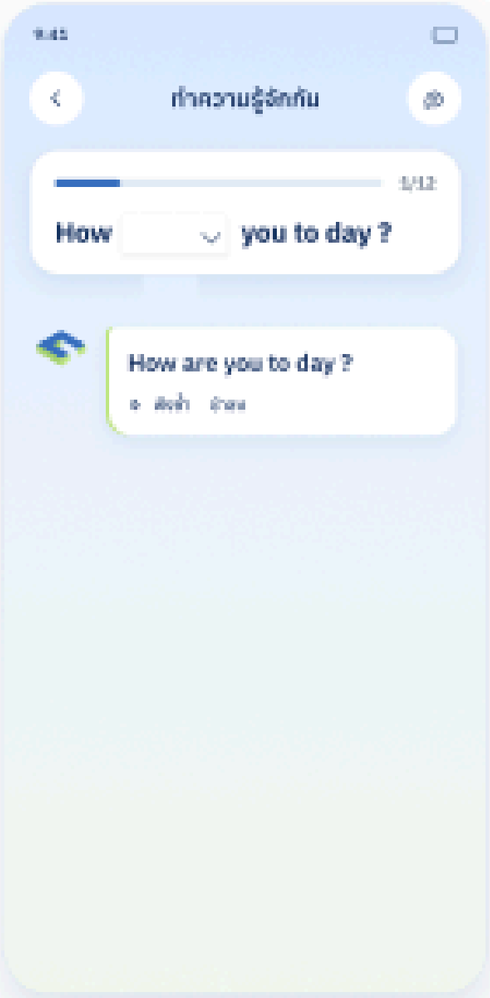
3 ตัวอย่าง Roleplay

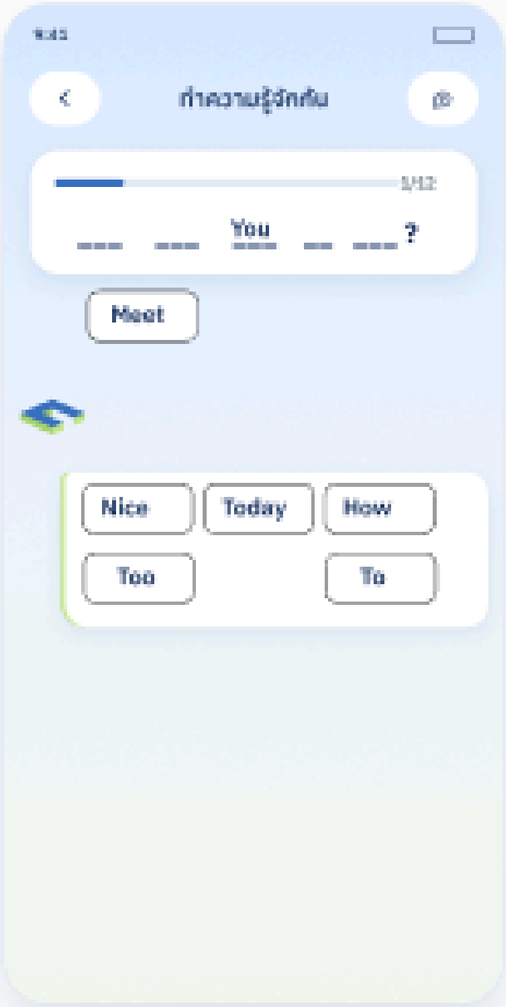




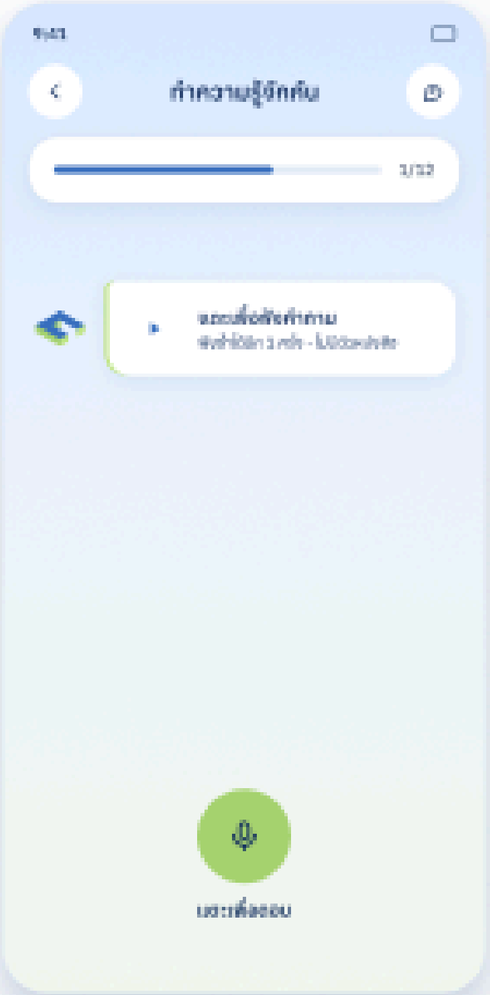


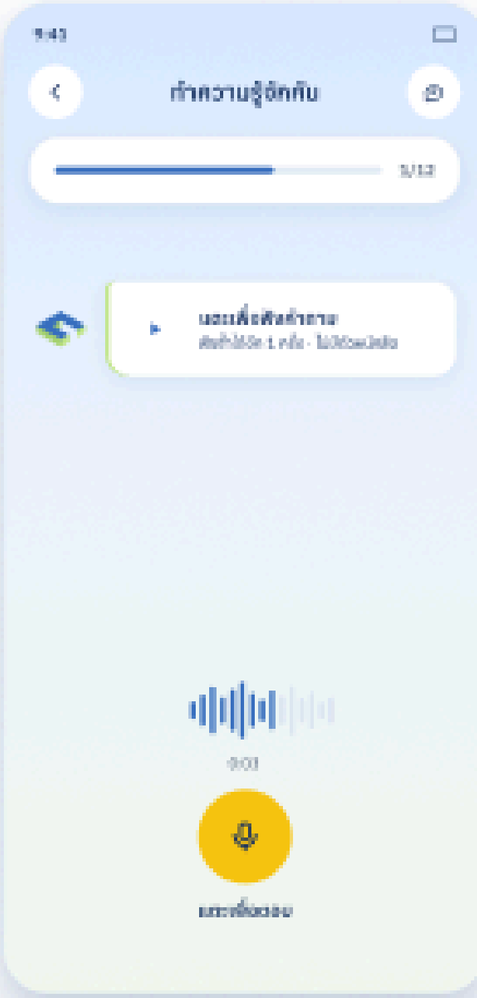
4 ตัวอย่าง แบบฝึกโวยครณ์



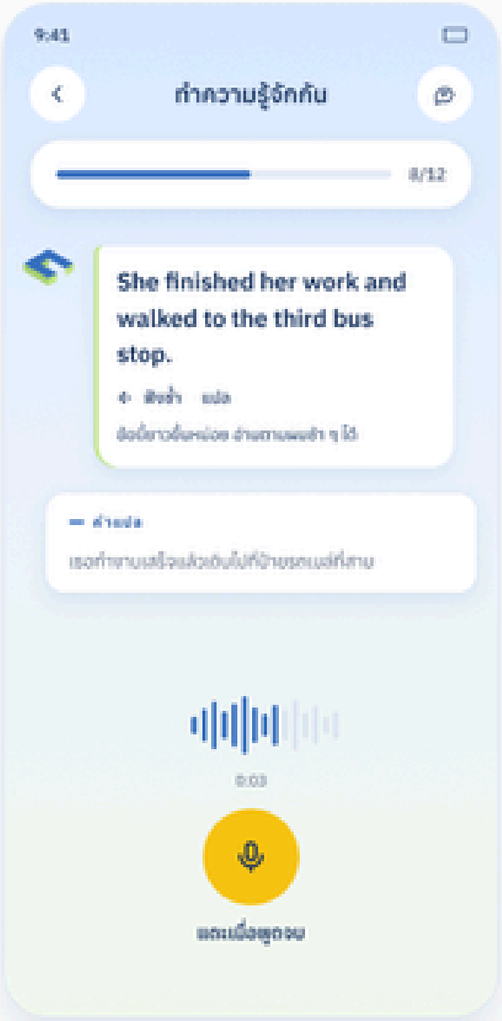


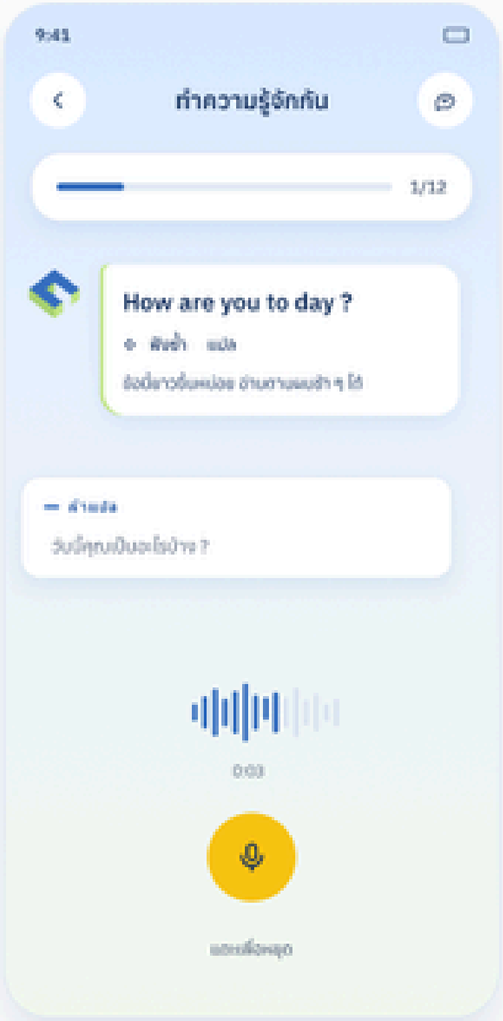
4 ตัวอย่าง แบบฝึกตรงเนื้อหา



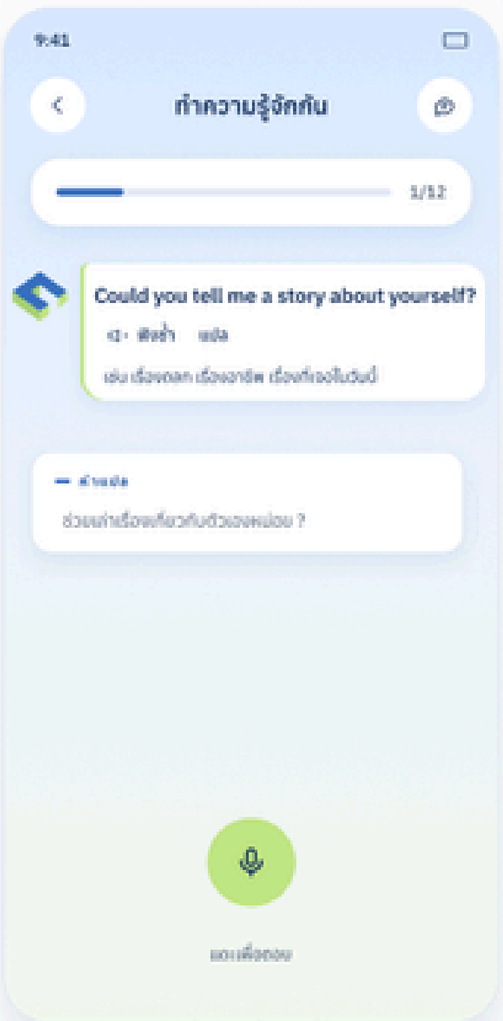


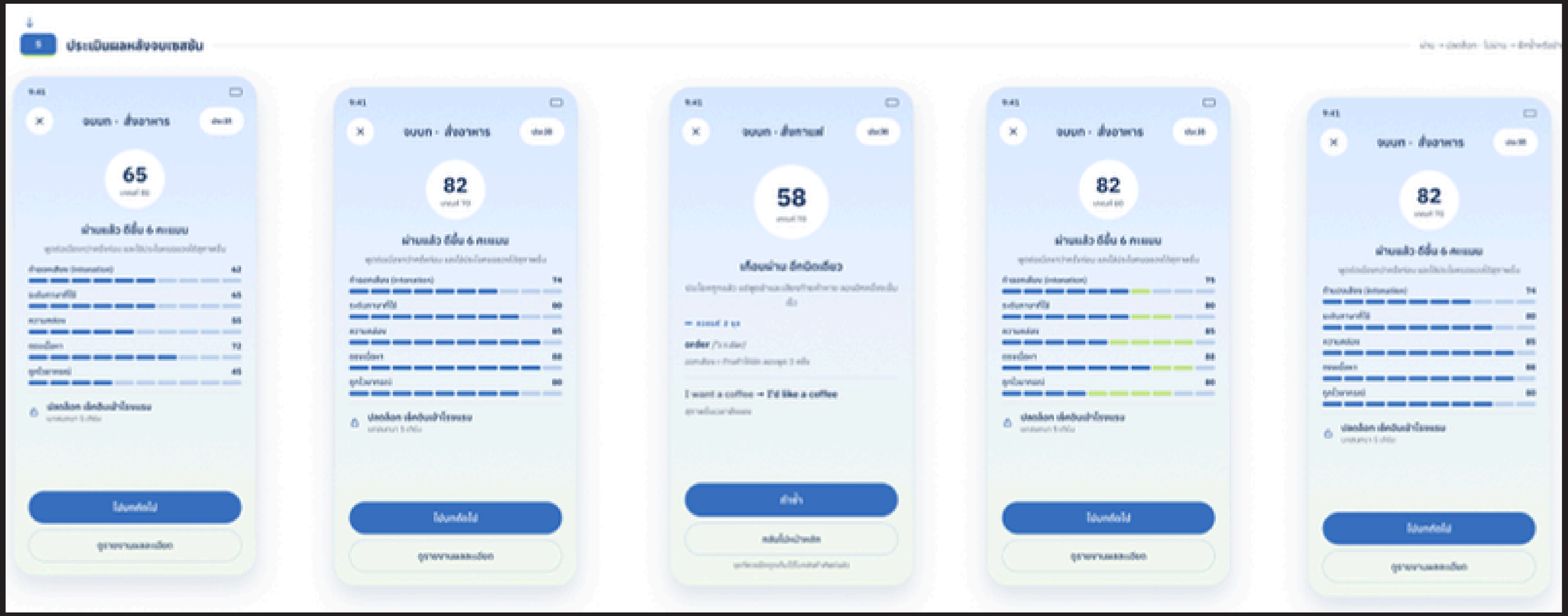
4 ตัวอย่าง แบบฝึกออกเสียง (intonation)



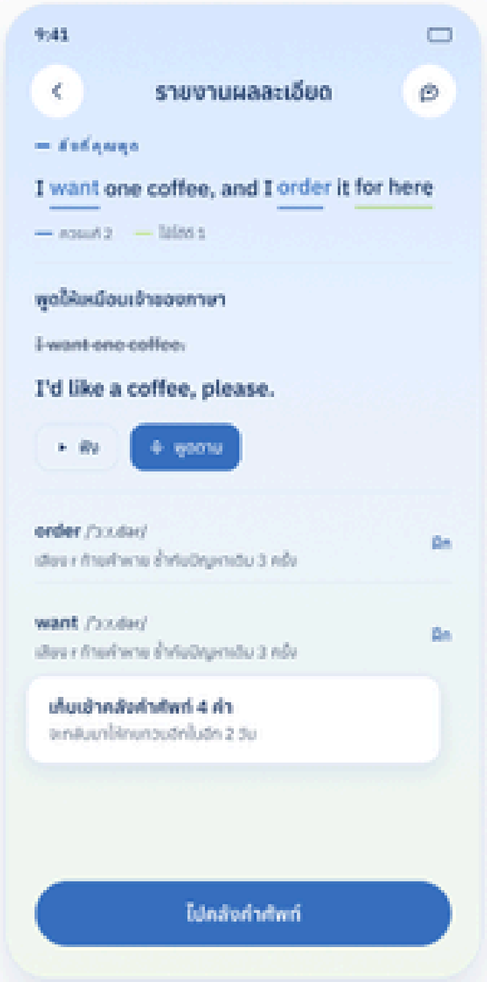


4 ตัวอย่าง แบบฝึกทดสอบความคล่อง

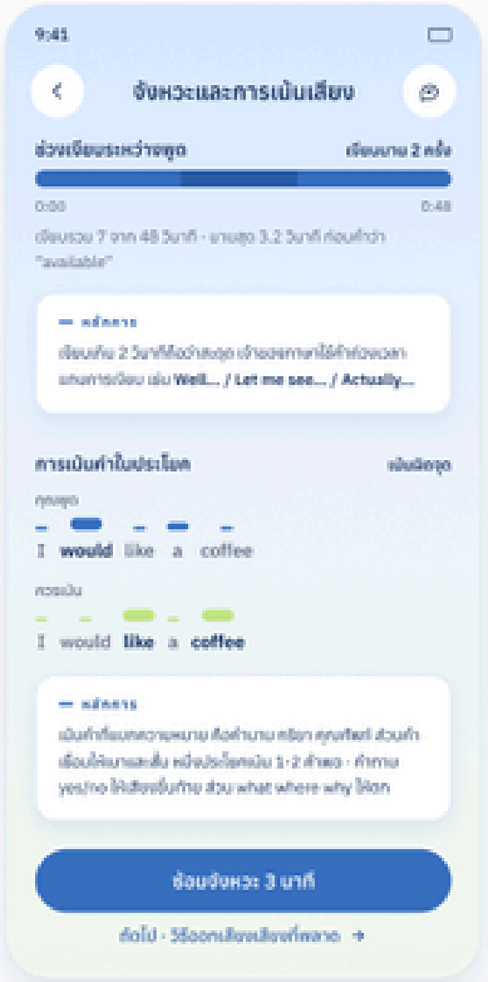



5
ประเมินผลหลังจบขั้นตอน
9:41
ขนม · สั่งอาหาร
65
ผ่านแล้ว ยังขึ้น 6 ทดสอบ
ไม่ผ่าน ยังต้องฝึกอีก
ดูรายงานผลฉบับเต็ม
9:41
ขนม · สั่งอาหาร
82
ผ่านแล้ว ยังขึ้น 6 ทดสอบ
ไม่ผ่าน ยังต้องฝึกอีก
ดูรายงานผลฉบับเต็ม
9:41
ขนม · สั่งกาแฟ
58
เกือบผ่าน นิดเดียวเอง
order
ลองใหม่
ดูรายงานผลฉบับเต็ม
9:41
ขนม · สั่งอาหาร
82
ผ่านแล้ว ยังขึ้น 6 ทดสอบ
ไม่ผ่าน ยังต้องฝึกอีก
ดูรายงานผลฉบับเต็ม
9:41
ขนม · สั่งอาหาร
82
ผ่านแล้ว ยังขึ้น 6 ทดสอบ
ไม่ผ่าน ยังต้องฝึกอีก
ดูรายงานผลฉบับเต็ม

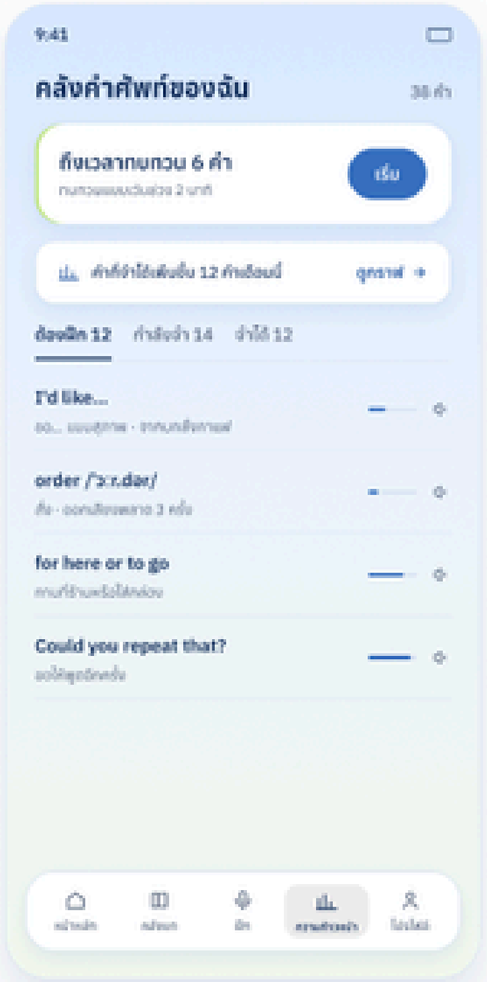
6 รายงานผลและคลังคำศัพท์



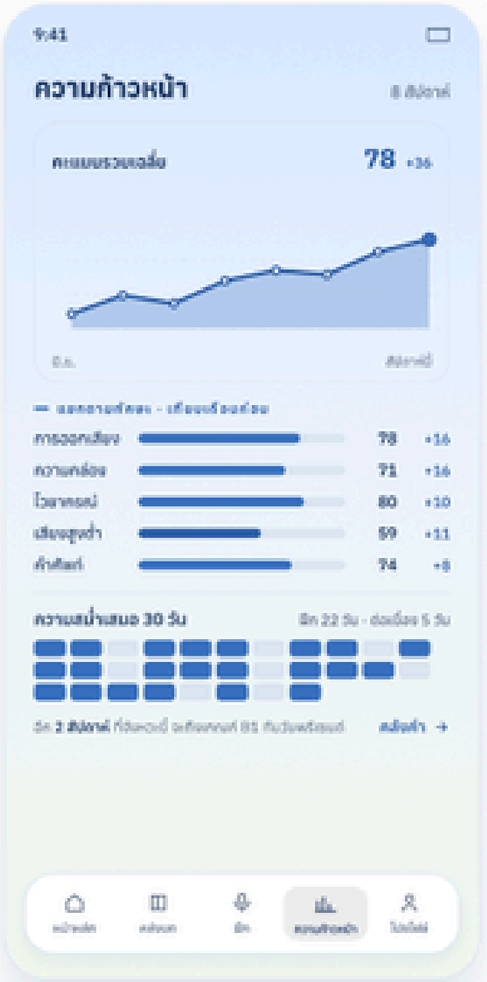
5.1 Report Card
ย้อนดูได้ทุกครั้ง + การเลือกที่จะฝึกใช้



5.2 จังหวะ ช่วงเงียบ และการเน้นเสียง
dead air + sentence stress + ทำนองเสียง พร้อมหลักการ



5.4 Personal Word Bank
เก็บอัตโนมัติ → spaced repetition



5.5 กราฟความก้าวหน้า
คะแนนรายสัปดาห์ + แยกทักษะ + ความสม่ำเสมอ